Aspect-Level Sentiment and Trend Analysis of Skincare Consumer Reviews

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The growing volume of online consumer reviews presents an opportunity for beauty brands to understand shifting consumer preferences and adapt their market strategy accordingly to remain competitive. Sentiment analysis enables brands to capture consumer reaction toward specific product aspects and inform growth strategies. However, existing approaches concentrate primarily on aggregate sentiment classification, with limited attention to aspect significance and temporal dynamics, which restricts the ability to track changes in consumer perception over time. This paper proposes a framework that addresses this limitation by integrating semantic chunking, topic modeling, taxonomy construction, and a cascaded aspect-based sentiment analysis (ABSA) classifier to extract aspect-level sentiment and facilitate trend analysis at year-level granularity. A systematic experiment comparing joint and cascaded architectures across multiple backbone models and learning paradigms validates the design decisions within the pipeline, establishing the cascaded architecture as the superior formulation with fine-tuned DeBERTa selected for aspect detection classifier and SetFit-RoBERTa for sentiment classifier. The framework is applied to 1.7 million skincare product reviews collected from Sephora over a decade. The advantage of the framework is demonstrated through its capability to surface temporal shifts in consumer concerns and sentiment dynamics across ten defined aspects that aggregate analysis would not capture.

Keywords: Aspect-based sentiment analysis, Temporal analysis, Text classification, Consumer reviews